

Based on 50 sources

Operator	Mathematical Domain	Application	Sources
Ξ (Recursive Identity Operator)	System Ontology / Higher Structure Theory	Stabilizes system identity through recursive self-negation and structural integrity audits; maps paradoxes to structural residues.	[1-4]
∂_H (Metamorphic Boundary Operator)	HALIRA Framework / Algebraic Topology	Acts as a Tier-0 boundary involution ($\partial_H^2 = id$) to ensure structural reversibility and stability in recursive research frameworks.	[5, 6]
Lang isogeny	Algebraic Geometry	Utilized within the Geometric Langlands Correspondence and character sheaf frameworks to extend results for connections with simple poles.	[5, U Desc
Cayley Transform	Deep Learning / Matrix Manifolds	Maps skew-symmetric matrices to the Stiefel manifold to ensure orthogonality and stabilize gradient flow in neural network hyper-connections.	[16, 1 Desc Infer
Sinkhorn Projections / Iteration	Deep Learning / Optimal Transport	Projects mixing matrices onto the Birkhoff polytope of doubly stochastic matrices to stabilize gradient flow and perform structural alignment.	[16, 1 Desc Infer
M (Molter Operator)	Algebraic Geometry / Physics	Utilizes torsion-memory tensor functions and chirality cascade logic to manage symbolic state collapse and define multiplicative local systems.	[1, 7]